

Prayas JEE 2.0 2027

Maths (GOAT)

Basic Mathematics

DPP: 8

Total Duration - 60 Min

Q1 If $\log_B A$ is defined then:

- (A) Both A and B should be positive
 (B) $A > 0$, $B > 0$ and $B \neq 1$
 (C) A and B both should be natural number
 (D) A or B can be negative

Q2 Complete set of values of x for which $\log_x(x+3)$ is always defined is

- (A) $x \in (-3, \infty)$
 (B) $x \in (0, \infty)$
 (C) $x \in (0, \infty) - \{1\}$
 (D) $x \in (-3, 0)$

Q3 If $\log_2(4 + \log_3(x)) = 3$, then sum of digits of x is

- (A) 3 (B) 6
 (C) 9 (D) 18

Q4 Sum of all the solution(s) of the equation $\log_{10}(x) + \log_{10}(x+2) - \log_{10}(5x+4) = 0$ is-

- (A) -1 (B) 3
 (C) 4 (D) 5

Q5 Solution set of $\sqrt{x^2 - 3x} \geq 2$ is

- (A) $x \in (-\infty, -1] \cup [4, \infty)$
 (B) $x \in (-\infty, 0] \cup [3, \infty)$
 (C) $x \in (-\infty, -1) \cup [4, \infty)$
 (D) None of these

Q6 Solution set of $x - 4 < \sqrt{x^2 + 4x - 12}$ is

- (A) $x \in [4, \infty)$
 (B) $x \in (-\infty, -6] \cup [2, \infty)$
 (C) $x \in R$
 (D) $x \in (-8, -3] \cup [6, \infty) \cup \{4\}$

Q7 Solution set of $(3^x - 9)(x^2 - 4)(7^x - 49)(5^x - 25) \geq 0$ is

- (A) $x \in [-2, \infty)$
 (B) $x \in (-\infty, -2) \cup (2, \infty)$

(C) $x \in [-2, 2]$ (D) $x \in (-\infty, 2]$ **Q8** Find the value of the Logarithm:

$$\log_5 20 + \log_5 \frac{125}{4} - \log_5 5^{-1}$$

Q9 Solve the logarithmic equation:

$$\log(3x^2 + 1) - \log(3 + x) = \log(3x - 2)$$

Q10 Solve the logarithmic equation: $3\log 2 - \log(x - 1)$

$$= \log(x + 1) - \log(x - 2)$$

Q11 Solve the logarithmic equation:

$$\log_3[1 + \log_3(2^x - 7)] = 1$$

Q12 Solve the logarithmic equation:

$$\log_7(2^x - 1) + \log_7(2^x - 7) = 1$$

Q13 Solve: $\log_{5-x}(x^2 - 2x + 65) = 2$ **Q14** Solve: $\log_7 \log_5(\sqrt{x+5} + \sqrt{x}) = 0$ **Q15** Solve: $\log_3(3^x - 8) = 2 - x$ **Q16** Solve: $\log_2(25^{x+3} - 1) = 2 + \log_2(5^{x+3} + 1)$ **Q17** Solve: $\log_2(4 \times 3^x - 6) - \log_2(9^x - 6) = 1$ **Q18** Solve: $\log_4(2 \times 4^{x-2} - 1) + 4 = 2x$ **Q19** If $\log(x - y) - \log 5 - \frac{1}{2}\log x - \frac{1}{2}\log y = 0$ then

$$\frac{x}{y} + \frac{y}{x} =$$

- (A) 25 (B) 26
 (C) 27 (D) 28

Q20 If $\log_3 2 + \log_3(2^x - 7/2) = 2\log_3(2^x - 5)$, then the value of x is

- (A) 3 (B) 2
 (C) 1 (D) 4



Q21 $\log_7 \log_7 \sqrt{7(\sqrt{7}\sqrt{7})} =$

- (A) $3 \log_2 7$
 (B) $1 - 3 \log_3 7$
 (C) $1 - 3 \log_7 2$
 (D) $1 - 10 \log_2 7$

Q22 The sum of all the solutions to the equation

$$2 \log_{10} x - \log_{10}(2x - 75) = 2$$

- (A) 30 (B) 350
 (C) 75 (D) 200

Q23

(a) If $\log_{10}(x^2 - 12x + 36) = 2$

(b) $9^{1+\log x} - 3^{1+\log x} - 210 = 0$; where base of log is 3.

Q24 If $2^{\log_{10} 3\sqrt{3}} = 3^{k \log_{10} 2}$, then $k =$

- (A) $\frac{1}{2}$
 (B) $\frac{3}{2}$
 (C) 3
 (D) 2



Answer Key

Q1 (B)
Q2 (C)
Q3 (C)
Q4 (C)
Q5 (A)
Q6 (B)
Q7 (A)
Q8 5
Q9 1
Q10 $x=3$ or 5
Q11 $x = 4$
Q12 $x = 3$

Q13 $x = -5$
Q14 $x = 4$
Q15 $x = 2$
Q16 $x = -2$
Q17 $x = 1$
Q18 $x = 2$
Q19 (C)
Q20 (A)
Q21 (C)
Q22 (D)
Q23 (a) 16, -4
(b) 5
Q24 (B)



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